reBot-DevArm B601-DM

ENGINEERING BUNDLE — Concept Phase

Architecture · Requirements · DFMEA · Test Plan · Safety Case · 2026-05-04 · github.com/jherrodthomas/reBot-DevArm

What's in this bundle. Consolidated view of the concept-phase engineering package for the open-source 6-DOF cobot arm. Detailed evidence in the companion workbooks. Provisional acceptance: OE-1 only.

Headline numbers

Hazards	Safety funcs	Requirements	DFMEA modes	High-AP	Test cases	Open gaps
33	31	104	27	11	131	28

1. Architecture (5 views)

Five complementary architectural views per ISO/IEC/IEEE 42010:2022. Each view derived from the upstream Seeed-Projects/reBot-DevArm repository and assumptions flagged.

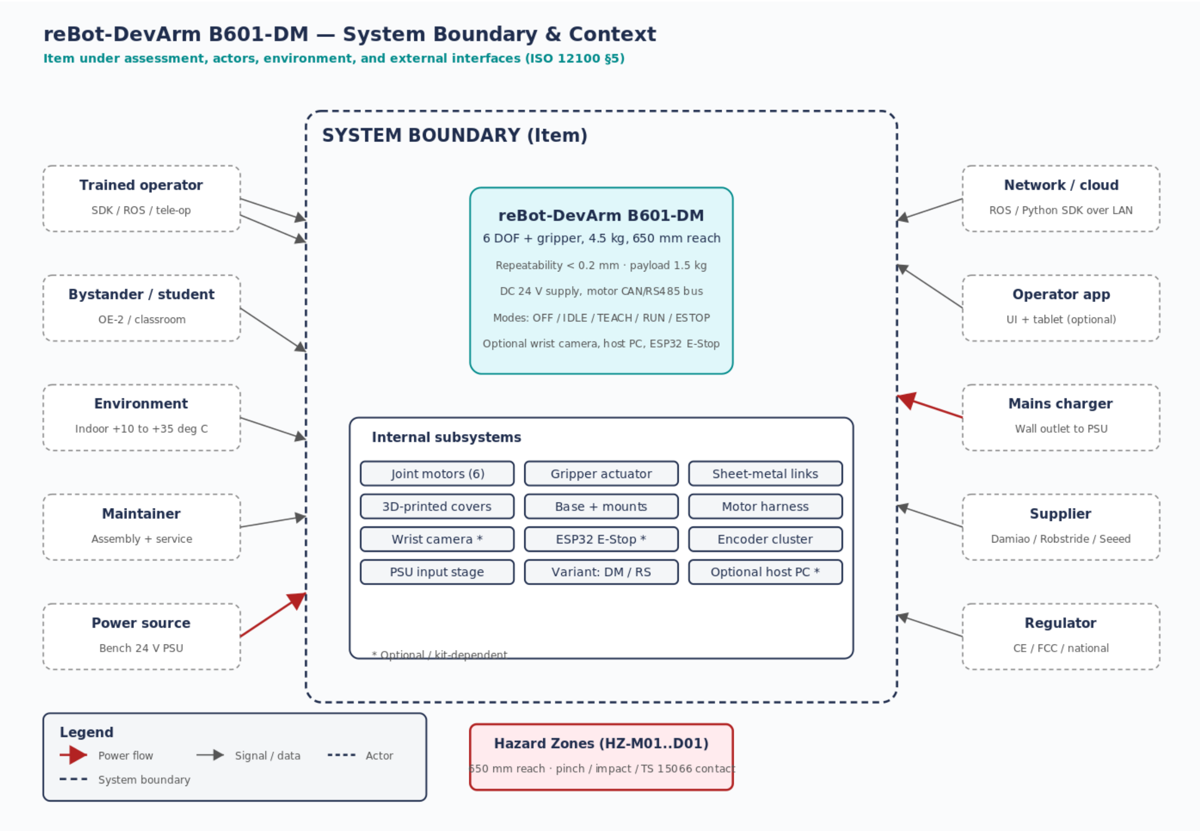

Figure 1 — System boundary & context (ISO 12100 §5).

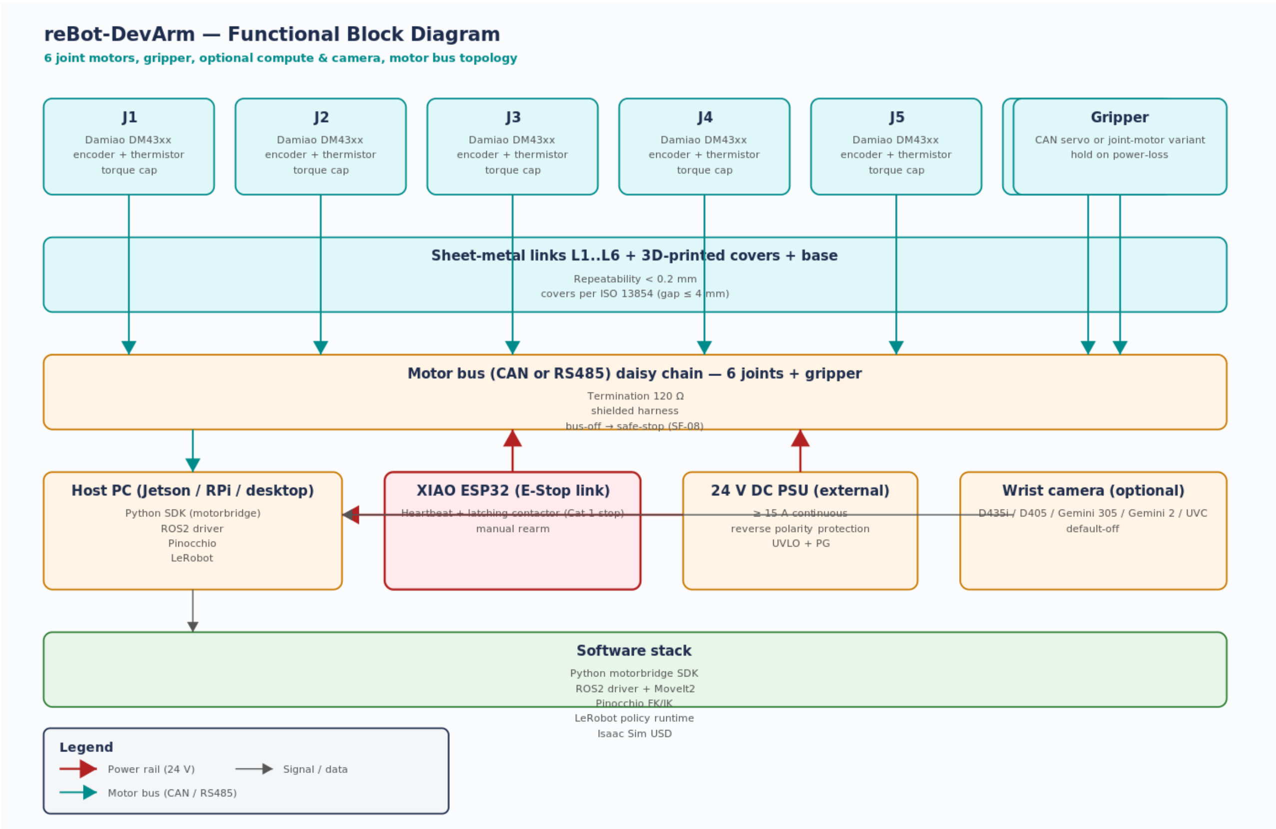

Figure 2 — Functional block diagram with motor bus, host, PSU, E-Stop.

Figure 3 — 31 safety functions allocated to HW (9), SW (15), Procedure (7).

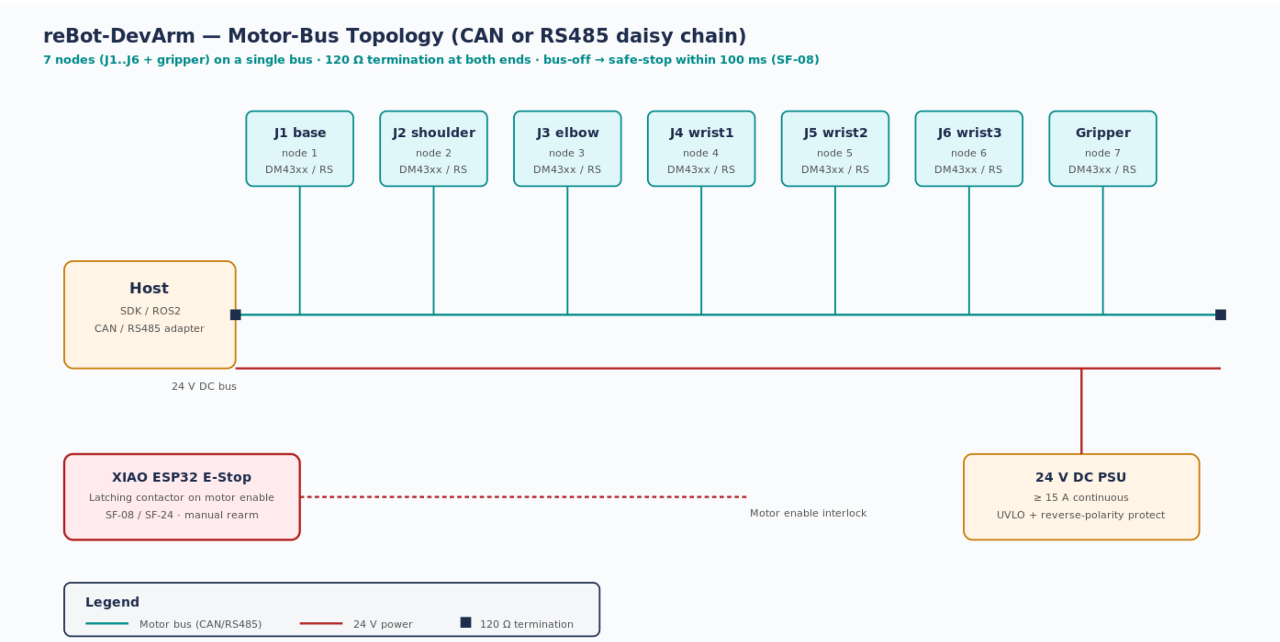

Figure 4 — Motor bus topology (CAN/RS485 daisy chain).

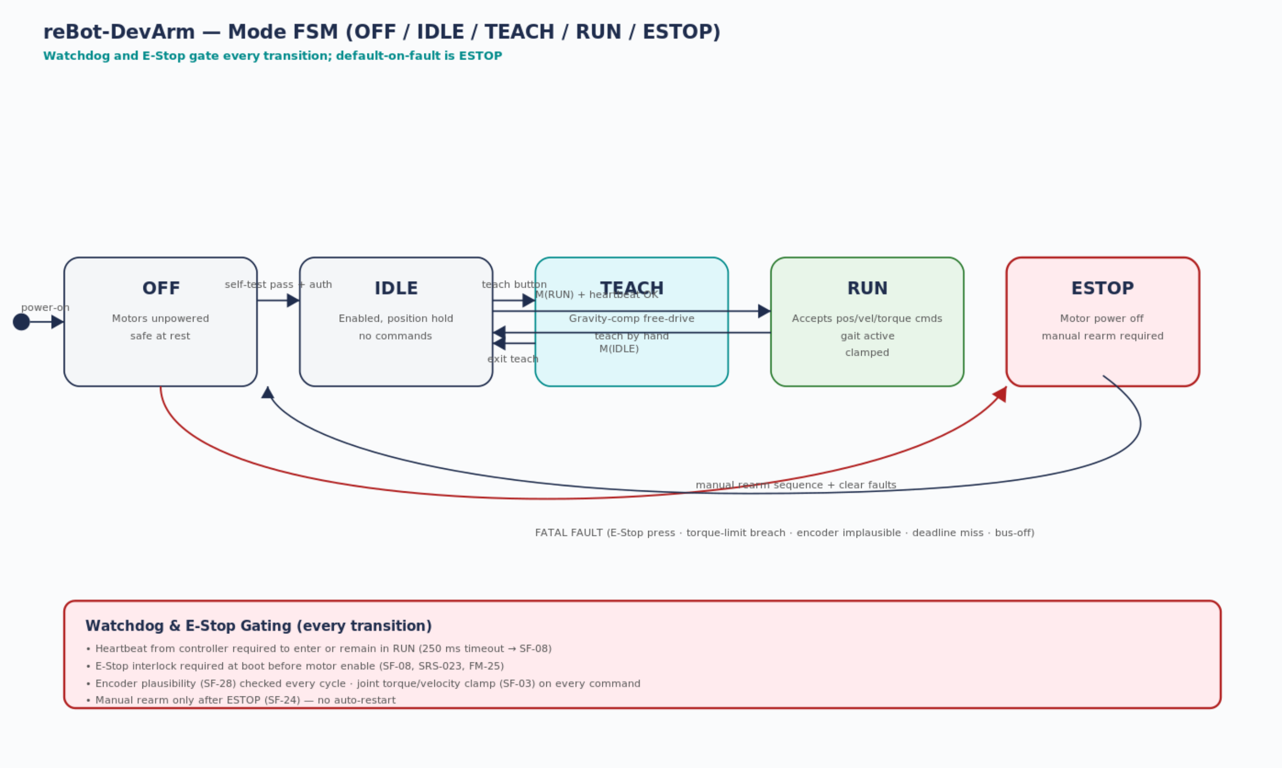

Figure 5 — Mode FSM (OFF / IDLE / TEACH / RUN / ESTOP).

2. Top safety requirements

REQ-ID	Type	Statement	SF	Tests
SYRS-002	SAFE	PFL per ISO/TS 15066	SF-01,SF-02	TC-S-002
SYRS-003	SAFE	Joint torque limit	SF-03	TC-U-001
SYRS-005	SAFE	Emergency stop ≤ 500 ms; fail-safe link loss	SF-08,SF-13	TC-S-004
SYRS-006	SAFE	Manual rearm after E-Stop	SF-24	TC-S-005
SYRS-008	SAFE	Watchdog safe-stop on comms loss	SF-08,SF-25	TC-S-006
SYRS-014	SAFE	Gripper hold on power loss	SF-13	TC-U-007
SYRS-026	SAFE	Camera default-off + privacy	SF-31	TC-Q-004
SYRS-029	SAFE	Pinch-zone gap ≤ 4 mm	SF-07	TC-U-015
HWRS-014	SAFE	E-Stop dual channel	SF-08	TC-S-041
HWRS-018	SAFE	Keyed PSU connector	SF-10	TC-U-059

3. Top High-AP DFMEA items

FM	Description	S	O	D	AP	Mitigation	Linked gap
FM-16	E-Stop button stuck / fails	10	3	4	H	Cat 3 dual channel	GAP-10218-02
FM-25	E-Stop wiring not specified / installed	10	4	3	H	Boot interlock with handshake	GAP-10218-02
FM-24	PFL not bound to joint torque caps	9	5	9	H	TS 15066 measurement	GAP-15066-01
FM-11	Motor bus → bus-off	8	4	4	H	Termination + safe-stop	GAP-13849-01
FM-01	Motor runaway	8	3	4	H	Cat 3 torque-monitor	GAP-13849-01
FM-18	SDK race / GIL stall	8	4	5	H	Deterministic core; watchdog	—
FM-20	Tele-op auth bypass	8	3	7	H	TLS+token+audit	GAP-CYB-01
FM-21	LeRobot policy edge case	8	5	7	H	Coverage matrix	GAP-LP-01
FM-27	Pinch-zone gap > 4 mm	8	3	7	H	Cover redesign	GAP-13854-01

Recommendation. Provisional acceptance for OE-1 only. Closing the top 10 gaps unlocks OE-1 unconditionally and OE-2 conditionally. OE-3/OE-4 require additional regulatory engagement.

Companion artefacts. reBot_DevArm_Safety_Case.xlsx · reBot_DevArm_Safety_Case_Report.docx · reBot_DevArm_Safety_Case_Summary.pdf · reBot_DevArm_DFMEA.xlsx · reBot_DevArm_Requirements.xlsx · reBot_DevArm_Architecture.docx · reBot_DevArm_TestPlan.xlsx · Diagrams/01..05 .svg + .png.